

Examples of Community Knowledge Guiding In-the-Wild Robot Deployments in Public Spaces

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	ROBOT (Technical & System)	HUMAN (Procedure, Researcher, Participant)	PUBLIC (Site, Authorities, Laws)
B E F O R E	Site and Robot Selection Robot Programming Site Visits Safe Robot Transportation to Site A priori Information and Resources In-Lab Pre-Testing Accessibility of Robot Features Site Pre-Testing Set Up Robot Infrastructure	Ethics: Consent and Compensation Data Protection Plan Data Storage Plan Preparation of Study Materials Study Procedure Data Collection and Auxiliary Equipment Team Training Researcher Support Plan	Seek Approval for Deployment Secure Access to Local Infrastructure Plan for Accessibility Inform the Public about the Deployment
D U R I N G	Robot Support Resetting the Experiment Documentation Robot Maintenance Robot Reassessment	Support for Members of the Public Researcher Support Data Protection Strategies Ensure Availability of Study Materials Documentation of the Deployment Verify Data Recording Dynamic Research Management	Maintain Contact with Stakeholders Risk Management Secure Accessibility and Safety Inform the Public
A F T E R	Robot Return Transportation Robot Infrastructure Teardown	Returning Equipment Team and Participant Compensation Document Initial Observations Long-Term Data Storage Data Post-Processing	Site Cleanup Engage with Local Authorities Engage Local Infrastructure Providers Inform the Public about the Results

Figure 1: The checklist used throughout this paper for robot deployments in public. Temporal phases (Before, During, After) are represented as table rows and domains (Robot, Human, Public) are represented as table columns. Topics to consider for each phase-domain combination are represented by yellow flip cards.

Abstract

Many of the challenges encountered in in-the-wild deployments remain undocumented despite there being many common pitfalls between different deployments. This creates a high barrier of entry and leads to the “HRI in the wild” community repeating avoidable mistakes. In this work, we discuss our own experiences using the

open source “HRI Public Deployment Checklist” website¹ and how learning from other researchers has had a direct impact on our own research strategies and design.

CCS Concepts

• Computer systems organization → Robotics; • Human-centered computing → Human computer interaction (HCI);

Keywords

Public spaces, Semi-public spaces, in-the-wild studies, checklist

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¹<https://hripublicdeploymentchecklist.github.io/>

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1 Introduction

In-the-wild studies operate in highly dynamic environments often largely outside the researcher's control. Each site is unique and thus there is no standardized practice for approaching studies in these spaces. These deployments are challenging, even for experienced researchers, and its challenges can never be fully anticipated prior to execution. Researchers have long shared anecdotes, insights and failures among each other at workshops and at academic conferences. We believe the shared knowledge accumulated as researchers communicate about their lessons learned is impactful for researchers at any stage of their "in-the-wild" study journeys.

In this paper, we provide concrete examples on how shared wisdom from researchers can improve other researchers' experiences. In our presented examples we use the "HRI public deployment checklist" [13] – hosted online² – from existing work [13]. This checklist contains knowledge from a diverse set of researchers with experience in in-the-wild deployments in public spaces.

We explore how access to other researchers' expertise through this "checklist" has guided each of our own studies personally. Our experiences serve as evidence of how shared community knowledge directly supports researchers in their study deployments and how experiences from others guides how we consider our own work going forward.

2 Background

2.1 Public In-the-Wild Deployments

Robots have been deployed in many types of public spaces such as museums [18, 23, 24], malls [3, 12], transportation hubs [1, 11, 28], universities [7, 14, 26, 27], city centers [2, 4] and sidewalks [6, 17, 19, 29]. Public robot deployments face unstructured environments where observed human-robot interactions range from typical to unusual [1, 16, 26]. For example, service robots in public might unexpectedly be perceived as unpredictable due to robot novelty or lack of communication behaviors [5, 15].

Anticipating and adapting to these unusual events is commonplace for researchers experienced with public HRI. The requisite knowledge is largely gained from personal experience or shared through word-of-mouth, with little being shared with the larger community in a structured manner.

2.2 Community Driven Best Practices

The HRI research community more broadly has long worked towards identifying best practices by providing community resources and guidance. Guidelines on useful HRI metrics [25] and questionnaires [21] have been practical, usable resources available to researchers. Additional primers on conducting empirical research

in Human-Robot Interaction [8], reporting for wizard-of-oz studies [20] and participatory design [30] enable junior researchers to learn from the experiences of the broader community.

Over the last few years, alongside this year's FailForward@HRI, workshops such as "Real-World HRI in Public and Private Spaces: Successes, Failures, and Lessons Learned" [10] and "Human-Robot Interaction in Extreme and Challenging Environments" [9] as well as our planned 2024 workshop "Abuse, Failures, Outliers - Mapping Unexpected Events in Human-Robot Interaction in the Wild"³ aim to create spaces where researchers can share anecdotes and insights demonstrate the community interest in knowledge sharing.

These events, as well as the development of resources, such as the community tool presented by Saad et al. [22] for other subareas in HRI, suggest an increasing demand for shared resources and continued dialogue.

Our recent checklist for public robot deployments [13] is, to our knowledge, the first to disseminate tacit knowledge specifically geared towards studying robots in public spaces. In this paper, we use this checklist as a case study to explore direct effects from the use of these kinds of community driven resources. We expect the benefits we have seen through the use of this public studies checklist will be observed from other resources for the broader "in-the-wild" community going forward and can serve as further motivation for our community to share our knowledge with each other.

3 The "Human Robot Interaction Public Deployment Checklist" and its Use in this Work

The checklist is a structured 3x3 table (Figure 1) consisting of two dimensions capturing the temporal phases and responsibility domains of a public robot deployment. Each cell contains multiple flip cards with a "topic" name on one side and a bullet point list of relevant corresponding "items". For instance, the checklist topic **Site Visit** should happen **Before** the deployment and is relevant to the domain of **Robot** preparation. "Site Visit" has checklist items such as **Explore how the robot will move in the space**, but also potential oversight for both junior and seasoned researchers – **Visit site at a similar time to expected deployment**. The existing pre-print details the development process and more of the checklist [13] features, including how it can be contributed to, personalized, and improved upon by the community through its github repository⁴. However, this paper focuses on the checklist used through the customization and "direct download as a Microsoft Word file" feature.

The checklist was developed over the course of three years from 2022 through 2025 but as it improves with community input and evolves through multiple versions, we must document that the case studies presented in this paper use the September 2025 version of the checklist.

²provided at <https://hripublicdeploymentchecklist.github.io/>

³unfortunately tabled due to unforeseeable circumstances

⁴<https://github.com/hripublicdeploymentchecklist>

4 Examples of Using the Checklist

In this work, we present the use of the “Human Robot Interaction Public Deployment Checklist” on three studies. Although the studies used the entire checklist as instructed, we will focus on a selection of specific checklist items that we found personally impactful on our respective studies. We include Claire’s use of the topic “Document Initial Observations” in the After+Human cell and “Researcher Support” from the During+Human cell; Sydney’s use of “Study Procedure” from Before+Human and “Documentation of the Deployment” from During+Human and Hannah’s and Franziska’s use of “Site Visit” and “Researcher Support Plan” from Before+Robot and Before+Human cell.

4.1 Claire

Claire’s personal account documents a study that is part of a long term participatory design initiative with the goal of long-term deployment of robots working with radiation safety teams for regular use in semi-public spaces. The study included collaboration with professionals in the radiation safety field at national labs, reactors, and medical environments. The research team is composed of three roboticists, two of which had not had experience with interaction work in the past.

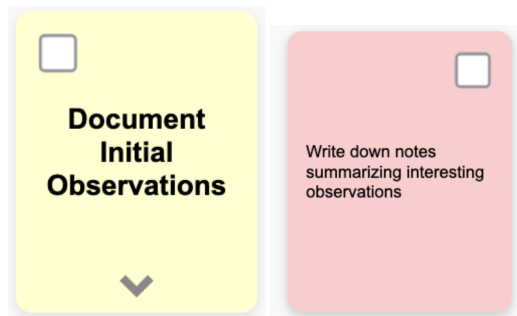


Figure 2: The “Document Initial Observations” topic card and checklist content included on the front and back.

Document Initial Observations: This topic, included in the After+Human cell, suggested that “it is useful to document initial observations based on the entirety of the deployment shortly after it finishes.” Claire built this documentation process into the study design itself such that during any interviews or collaboration sessions with experts, said initial observations were automatically stored for later analysis without necessary retroactive documentation. In particular, observations on the attitudes radiation professionals had towards speculative robot systems. These “initial observation” notes influenced later participatory design stages. Without anticipation of this need from the checklist, the later co-design of robot form study would have been missing key insights.

Researcher Support: From the “researcher support” topic in cell During+Human, Claire drew from the checklist point to “Have an open communication with (in-) formal team members” and “have regular debriefing meetings.” As a result, she set regular meetings with the entire team as well as an asynchronous communication

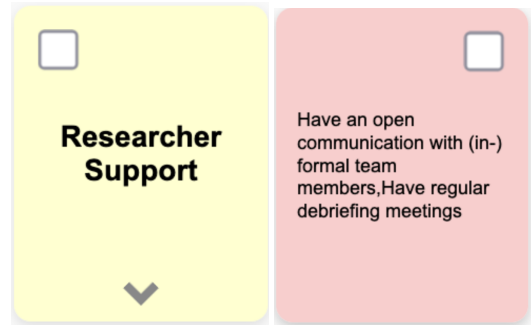
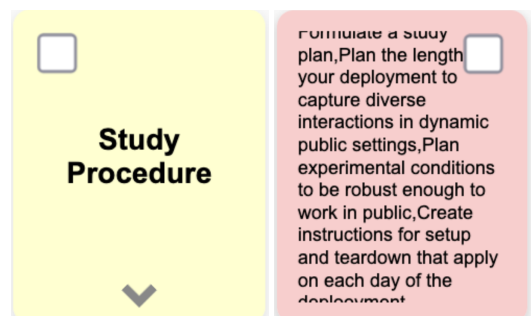


Figure 3: The topic card “Researcher Support” from the During+Human cell and checklist content included on the front and back.

channel. By having fixed regular meetings rather than more irregular ad-hoc strategies she had taken in the past, the team communication we much more integrated than some of Claire’s prior studies and the collaborative nature of the team impacted the quality of their findings.



Study Procedure

Checklist Items

- Formulate a study plan
- Plan the length of your deployment to capture diverse interactions in dynamic public settings
- Plan experimental conditions to be robust enough to work in public
- Create instructions for setup and teardown that apply on each day of the deployment

Figure 4: The “study procedure” topic card and checklist content from the checklist included on the front and back as well as the full expanded text on the checklist items below.

4.2 Sydney

Sydney conducted a robot deployment in a publicly accessible rotunda at a university. They had deployed the same robot multiple times previously, but were at a new location with new handlers.

Study Procedure: This topic inspired Sydney to create “how to start the robot” and “what to do when things go wrong” guides. These guides were used by robot handlers when performing troubleshooting on-site without Sydney’s direct supervision and were directly beneficial to the study execution. Without those guides, the different handlers across many sessions would have been much more prone to error due to the complexity of the deployment.

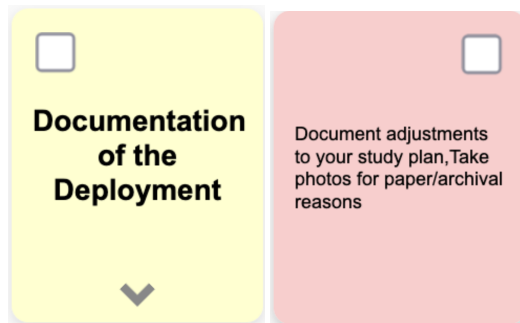


Figure 5: The topic card “Documentation of the Deployment” from the During+Human cell and checklist content from the checklist included on the front and back.

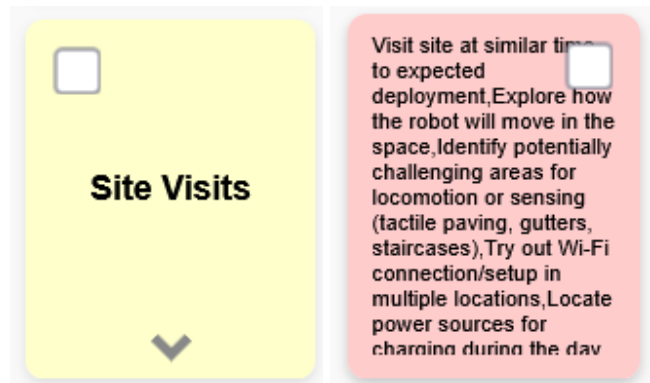
Documentation of the Deployment: This topic, much like Claire’s experience, suggested robot handlers to document interesting observations throughout the deployment. One such observation was the effects of weather changes on human behavior in this new deployment site. This major change in human behavior was an important context factor to note for future use of the dataset. Without the reminder from the checklist for proper documentation this might not have been noted.

4.3 Hannah and Franziska

Hannah and Franziska deployed a mobile robot in a student library together with a robot handler and a student operating the camera to film the events. While nearly all checklist items were relevant for their deployment this case study focuses on “Site Visits” and “Researcher Support Plan”.

Site Visits: For this deployment, the researchers performed a site visit to check if the student library common area sufficiently accommodated the robot (e.g., location of stairs and floor markings). As a result, the route planned was able to avoid areas where the robot could get stuck while simultaneously allowing a route where human interaction could be observed.

Researcher Support Plan: The first deployment team initially consisted of four people. Later on, the second deployment team included two more students to obtain consent from filmed participants. The “researcher support plan” checklist recommendations encouraged this addition to alleviate workload on Hannah and Franziska as during the first deployment it was difficult for only two people to manage all of the needed tasks.



Site Visits

Checklist Items

- Visit site at similar time to expected deployment
- Explore how the robot will move in the space
- Identify potentially challenging areas for locomotion or sensing (tactile paving, gutters, staircases)
- Try out Wi-Fi connection/setup in multiple locations
- Locate power sources for charging during the day

Figure 6: The “Site Visit” topic card from the Before+Robot cell and checklist content from the checklist included on the front and back, as well as bullet point items visible as expanded text.

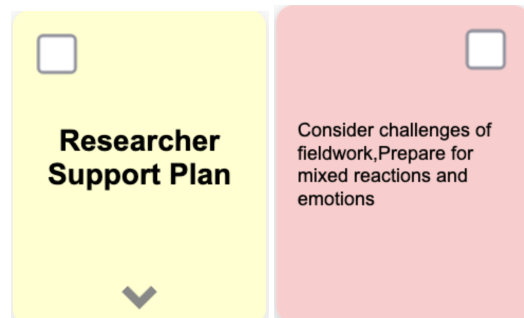


Figure 7: The topic card “Researchers Support Plan” from the Before+Human cell and checklist content from the checklist included on the front and back.

5 Discussion and Conclusion

In this paper, we present our own stories of how a community checklist synthesizing joint researchers’ knowledge can streamline deployment processes. It is an encouraging demonstration of how beneficial community-created resources such as the “Human Robot Interaction Public Deployment Checklist” can be. The checklist is open source and, as such, caters well to the “cause and effect” structure for the examples included in this paper. Although, the “checklist” is designed for public and semi-public deployments, its

lessons can be helpful for other in-the-wild studies. We see this checklist as a naturally evolving community research and look forward to including more topics based on shared experiences with in-the-wild studies.

The case studies using the “Human Robot Interaction Checklist” only highlight one method for sharing researcher knowledge. Although not as easily analyzed using the “cause and effect” structure in this work, valuable in-person dialogue at workshops such as Fail-Forward@HRI with open, candid, and constructive conversations only accessible in these gatherings certainly have similar impacts. These discussions and connections between researchers have a different form of long-reaching effects and would be interesting to explore and analyze the downstream benefits they serve junior researchers and future studies.

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